Adding storage capacity

StorageGRID 11.5

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Adding storage capacity

When existing Storage Nodes become full, you must increase the storage capacity of your StorageGRID system.

To increase storage capacity, you must first understand where data is stored currently and then add capacity in all required locations. For example, if you currently store copies of object data at several sites, you might need to increase the storage capacity of each site.

- Guidelines for adding object capacity
- Adding storage capacity for replicated objects
- Adding storage capacity for erasure-coded objects
- Considerations for rebalancing erasure-coded data

Guidelines for adding object capacity

You can expand the object storage capacity of your StorageGRID system by adding storage volumes to existing Storage Nodes or by adding new Storage Nodes to existing sites. You must add storage capacity in a way that meets the requirements of your information lifecycle management (ILM) policy.

Guidelines for adding storage volumes

Before adding storage volumes to existing Storage Nodes, review the following guidelines and limitations:

- You must examine your current ILM rules to determine where and when to add storage volumes to increase the storage available for replicated or erasure-coded objects. See the instructions for managing objects with information lifecycle management.
- You cannot increase the metadata capacity of your system by adding storage volumes because object metadata is stored only on volume 0.
- Each software-based Storage Node can support a maximum of 16 storage volumes. If you need to add capacity beyond that, you must add new Storage Nodes.
- You can add one or two expansion shelves to each SG6060 appliance. Each expansion shelf adds 16 storage volumes. With both expansion shelves installed, the SG6060 can support a total of 48 storage volumes.
- You cannot add storage volumes to any other storage appliance.
- You cannot increase the size of an existing storage volume.
- You cannot add storage volumes to a Storage Node at the same time you are performing a system upgrade, recovery operation, or another expansion.

After you have decided to add storage volumes and have determined which Storage Nodes you must expand to satisfy your ILM policy, follow the instructions for your type of Storage Node:

- To add expansion shelves to an SG6060 storage appliance, see the instructions for SG6000 appliance installation and maintenance.

  SG6000 storage appliances

- For a software-based node, follow the instructions for adding storage volumes to Storage Nodes.

  Adding storage volumes to Storage Nodes

Guidelines for adding Storage Nodes

Before adding Storage Nodes to existing sites, review the following guidelines and limitations:

- You must examine your current ILM rules to determine where and when to add Storage Nodes to increase the storage available for replicated or erasure-coded objects.
- You should not add more than 10 Storage Nodes in a single expansion procedure.
- You can add Storage Nodes to more than one site in a single expansion procedure.
- You can add Storage Nodes and other types of nodes in a single expansion procedure.
- Before starting the expansion procedure, you must confirm that all data-repair operations performed as part of a recovery are complete. See the steps for checking data repair jobs in the recovery and maintenance instructions.
- If you need to remove Storage Nodes before or after performing an expansion, you should not decommission more than 10 Storage Nodes in a single Decommission Node procedure.

Guidelines for the ADC service on Storage Nodes

When configuring the expansion, you must choose whether to include the Administrative Domain Controller (ADC) service on each new Storage Node. The ADC service keeps track of the location and availability of grid services.

- The StorageGRID system requires a quorum of ADC services to be available at each site and at all times.

  Learn about the ADC quorum in the recovery and maintenance instructions.

- At least three Storage Nodes at each site must include the ADC service.
- Adding the ADC service to every Storage Node is not recommended. Including too many ADC services can cause slowdowns due to the increased amount of communication between nodes.
- A single grid should not have more than 48 Storage Nodes with the ADC service. This is equivalent to 16 sites with three ADC services at each site.
- In general, when you select the **ADC Service** setting for a new node, you should select **Automatic**. Select **Yes** only if the new node will replace another Storage Node that includes the ADC service. Because you cannot decommission a Storage Node if too few ADC services would remain, this ensures that a new ADC service is available before the old service is removed.
- You cannot add the ADC service to a node after it is deployed.

Related information

Manage objects with ILM

SG6000 storage appliances

Adding storage volumes to Storage Nodes

Maintain & recover

Adding storage capacity for replicated objects

If the information lifecycle management (ILM) policy for your deployment includes a rule that creates replicated copies of objects, you must consider how much storage to add and where to add the new storage volumes or Storage Nodes.

For guidance on where to add additional storage, examine the ILM rules that create replicated copies. If ILM rules create two or more object copies, plan to add storage in each location where object copies are made. As a simple example, if you have a two-site grid and an ILM rule that creates one object copy at each site, you must add storage to each site to increase the overall object capacity of the grid.

For performance reasons, you should attempt to keep storage capacity and compute power balanced across sites. So, for this example, you should add the same number of Storage Nodes to each site or additional storage volumes at each site.

If you have a more complex ILM policy that includes rules that place objects in different locations based on criteria such as bucket name, or rules that change object locations over time, your analysis of where storage is required for the expansion will be similar, but more complex.

Charting how quickly overall storage capacity is being consumed can help you understand how much storage to add in the expansion, and when the additional storage space will be required. You can use the Grid Manager to monitor and chart storage capacity as described in the instructions for monitoring and troubleshooting StorageGRID.

When planning the timing of an expansion, remember to consider how long it might take to procure and install additional storage.

Related information

Manage objects with ILM

Monitor & troubleshoot

Adding storage capacity for erasure-coded objects

If your ILM policy includes a rule that makes erasure-coded copies, you must plan where to add new storage and when to add new storage. The amount of storage you add and the timing of the addition can affect the grid's usable storage capacity.

The first step in planning a storage expansion is to examine the rules in your ILM policy that create erasure-coded objects. Because StorageGRID creates *k+m* fragments for every erasure-coded object and stores each fragment on a different Storage Node, you must ensure that at least *k+m* Storage Nodes have space for new erasure-coded data after the expansion. If the erasure-coding profile provides site-loss protection, you must add storage to each site.

The number of nodes you need to add also depends on how full the existing nodes are when you perform the expansion.

General recommendation for adding storage capacity for erasure-coded objects

If you want to avoid detailed calculations, you can add two Storage Nodes per site when existing Storage Nodes reach 70% capacity.

This general recommendation provides reasonable results across a wide range of erasure-coding schemes for both single-site grids and for grids where erasure coding provides site-loss protection.

To better understand the factors that lead to this recommendation or to develop a more precise plan for your site, review the next section. For a custom recommendation optimized for your situation, contact your NetApp account representative.

Calculating the number of expansion Storage Nodes to add for erasure-coded objects

To optimize how you expand a deployment that stores erasure-coded objects, you must consider many factors:

- Erasure-coding scheme in use
- Characteristics of the storage pool used for erasure coding, including the number of nodes at each site and the amount of free space on each node
- Whether the grid was previously expanded (because the amount of free space per Storage Node might not be approximately the same on all nodes)
- Exact nature of the ILM policy, such as whether ILM rules make both replicated and erasure-coded objects

The following examples can help you understand the impact of the erasure-coding scheme, the number of nodes in the storage pool, and the amount of free space on each node.

Similar considerations affect the calculations for an ILM policy that stores both replicated and erasure-coded data and the calculations for a grid that has been previously expanded.

> The examples in this section represent the best practices for adding storage capacity to a StorageGRID system. If you are unable to add the recommended number of nodes, you might need to run the EC rebalance procedure to allow additional erasure-coded objects to be stored.
>
> Considerations for rebalancing erasure-coded data

Example 1: Expanding a one-site grid that uses 2+1 erasure coding

This example shows how to expand a simple grid that includes only three Storage Nodes.

> This example uses only three Storage Nodes for simplicity. However, using only three Storage Nodes is not recommended: an actual production grid should use a minimum of $k+m + 1$ Storage Nodes for redundancy, which equals four Storage Nodes (2+1+1) for this example.

Assume the following:

- All data is stored using the 2+1 erasure-coding scheme. With the 2+1 erasure coding scheme, every object is stored as three fragments, and each fragment is saved on a different Storage Node.
- You have one site with three Storage Nodes. Each Storage Node has a total capacity of 100 TB.
- You want to expand by adding new 100 TB Storage Nodes.

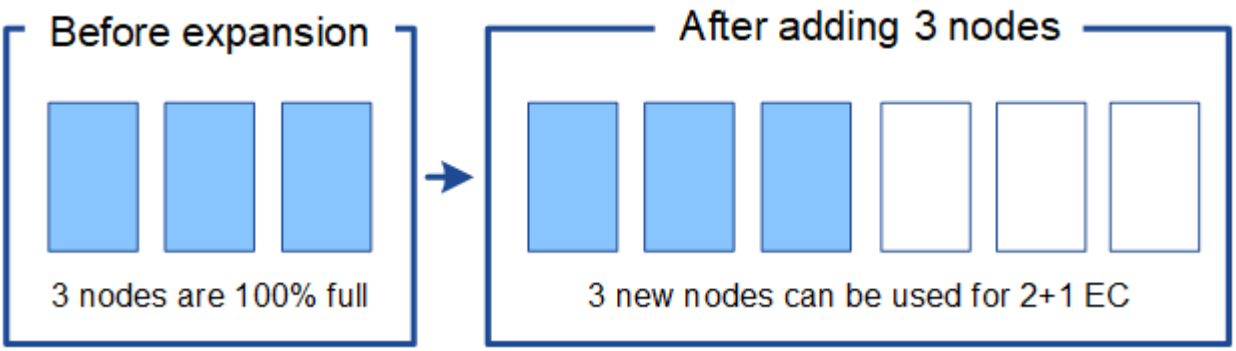
- You want to eventually balance erasure-coded data across the old and new nodes.

You have a number of options, based on how full the Storage Nodes are when you perform the expansion.

- **Add three 100 TB Storage Nodes when the existing nodes are 100% full**

  In this example, the existing nodes are 100% full. Because there is no free capacity, you must immediately add three nodes to continue 2+1 erasure coding.

  After the expansion is complete, when objects are erasure-coded, all fragments will be placed on the new nodes.

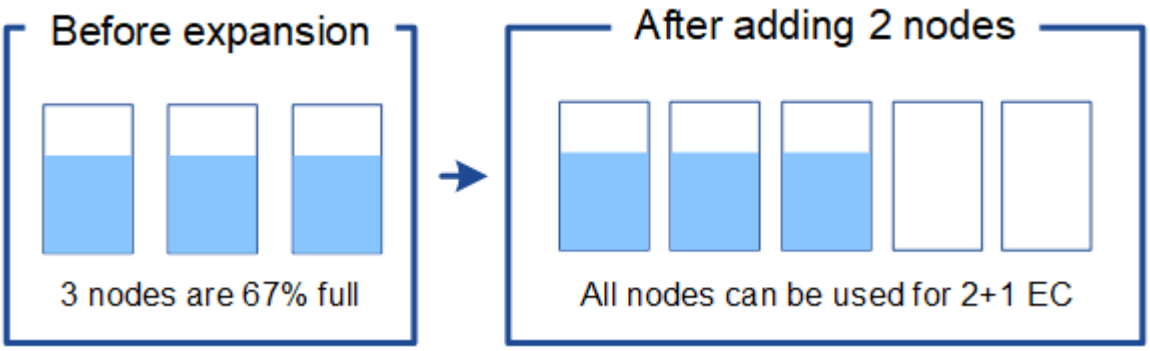


> This expansion adds *k+m* nodes. Adding four nodes is recommended for redundancy. If you add only *k+m* expansion Storage Nodes when existing nodes are 100% full, all new objects must be stored on the expansion nodes. If any of the new nodes become unavailable, even temporarily, StorageGRID cannot meet ILM requirements.

- **Add two 100 TB Storage Nodes, when the existing Storage Nodes are 67% full**

  In this example, the existing nodes are 67% full. Because there are 100 TB of free capacity on the existing nodes (33 TB per node), you only need to add two nodes if you perform the expansion now.

  Adding 200 TB of additional capacity will allow you to continue 2+1 erasure coding and to eventually balance erasure-coded data across all nodes.

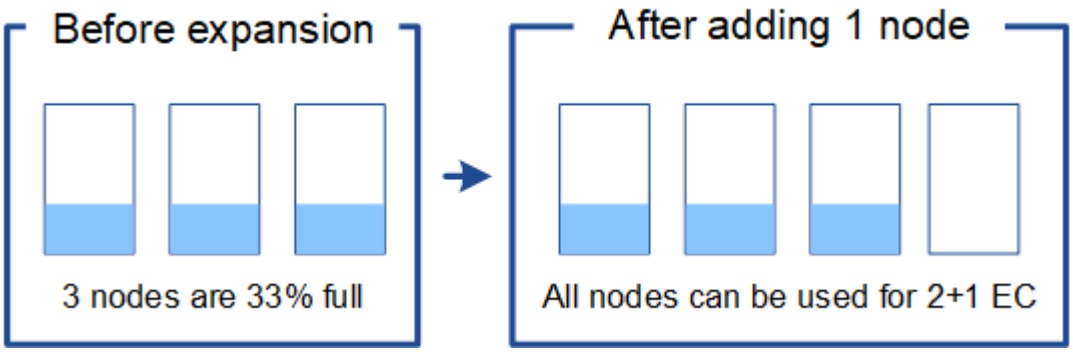


- **Add one 100 TB Storage Node when the existing Storage Nodes are 33% full**

  In this example, the existing nodes are 33% full. Because there are 200 TB of free capacity on the existing nodes (67 TB per node), you only need to add one node if you perform the expansion now.

  Adding 100 TB of additional capacity will allow you to continue 2+1 erasure coding and to eventually

balance erasure-coded data across all nodes.



Example 2: Expanding a three-site grid that uses 6+3 erasure coding

This example shows how to develop an expansion plan for a multi-site grid that has an erasure-coding scheme with a larger number of fragments. Despite the differences between these examples, the recommended expansion plan is very similar.

Assume the following:

- All data is stored using the 6+3 erasure coding scheme. With the 6+3 erasure coding scheme, every object is stored as 9 fragments, and each fragment is saved to a different Storage Node.
- You have three sites, and each site has four Storage Nodes (12 nodes in total). Each node has a total capacity of 100 TB.
- You want to expand by adding new 100 TB Storage Nodes.
- You want to eventually balance erasure-coded data across the old and new nodes.

You have a number of options, based on how full the Storage Nodes are when you perform the expansion.

- **Add nine 100 TB Storage Nodes (three per site), when existing nodes are 100% full**

  In this example, the 12 existing nodes are 100% full. Because there is no free capacity, you must immediately add nine nodes (900 TB of additional capacity) to continue 6+3 erasure coding.

  After the expansion is complete, when objects are erasure-coded, all fragments will be placed on the new nodes.

  > This expansion adds *k+m* nodes. Adding 12 nodes (four per site) is recommended for redundancy. If you add only *k+m* expansion Storage Nodes when existing nodes are 100% full, all new objects must be stored on the expansion nodes. If any of the new nodes become unavailable, even temporarily, StorageGRID cannot meet ILM requirements.

- **Add six 100 TB Storage Nodes (two per site), when existing nodes are 75% full**

  In this example, the 12 existing nodes are 75% full. Because there are 300 TB of free capacity (25 TB per node), you only need to add six nodes if you perform the expansion now. You would add two nodes to each of the three sites.

  Adding 600 TB of storage capacity will allow you to continue 6+3 erasure coding and to eventually balance erasure-coded data across all nodes.

- **Add three 100 TB Storage Nodes (one per site), when existing nodes are 50% full**

  In this example, the 12 existing nodes are 50% full. Because there are 600 TB of free capacity (50 TB per node), you only need to add three nodes if you perform the expansion now. You would add one node to each of the three sites.

  Adding 300 TB of storage capacity will allow you to continue 6+3 erasure coding and to eventually balance erasure-coded data across all nodes.

Related information

Manage objects with ILM

Monitor & troubleshoot

Considerations for rebalancing erasure-coded data

Considerations for rebalancing erasure-coded data

If you are performing an expansion to add Storage Nodes and your ILM policy includes one or more ILM rules to erasure code data, you might need to perform the EC rebalance procedure after the expansion is complete.

For example, if you cannot add the recommended number of Storage Nodes in an expansion, you might need to run the EC rebalance procedure to allow additional erasure-coded objects to be stored.

What is EC rebalancing?

EC rebalancing is a StorageGRID procedure that might be required after a Storage Node expansion. The procedure is run as a command-line script from the primary Admin Node. When you run the EC rebalance procedure, StorageGRID redistributes erasure-coded fragments among the existing and the newly expanded Storage Nodes at a site.

When the EC rebalance procedure runs:

- It only moves erasure-coded object data. It does not move replicated object data.
- It redistributes the data within a site. It does not move data between sites.
- It redistributes data among all Storage Nodes at a site. It does not redistribute data within storage volumes.

When the EC rebalance procedure is complete:

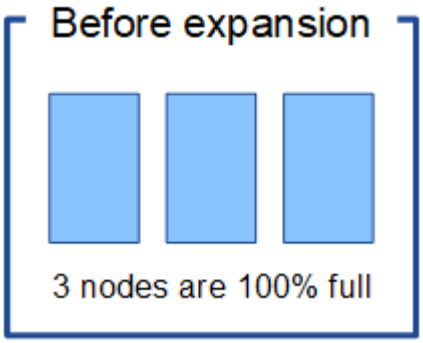
- Erasure-coded data is moved from Storage Nodes with less available space to Storage Nodes with more available space.
- Used (%) values might remain different between Storage Nodes because the EC rebalance procedure does not move replicated object copies.
- The data protection of erasure-coded objects will be unchanged.

When the EC rebalance procedure is running, the performance of ILM operations and S3 and Swift client operations are likely to be impacted. For this reason, you should only perform this procedure in limited cases.

When not to perform an EC rebalance

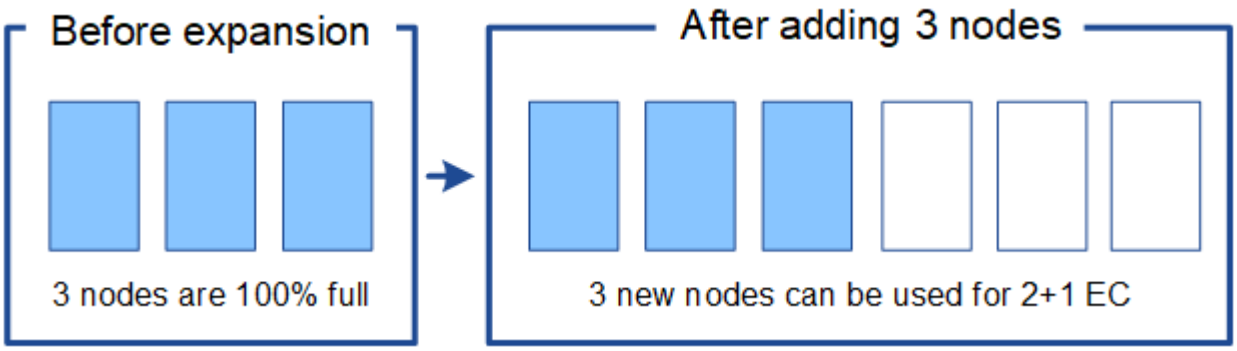
As an example of when you do not need to perform an EC rebalance, consider the following:

- StorageGRID is running at a single site, which contains three Storage Nodes.
- The ILM policy uses a 2+1 erasure-coding rule for all objects larger than 0.2 MB and a 2-copy replication rule for smaller objects.
- All Storage Nodes have become completely full, and the **Low Object Storage** alert has been triggered at the major severity level. The recommended action is to perform an expansion procedure to add Storage Nodes.



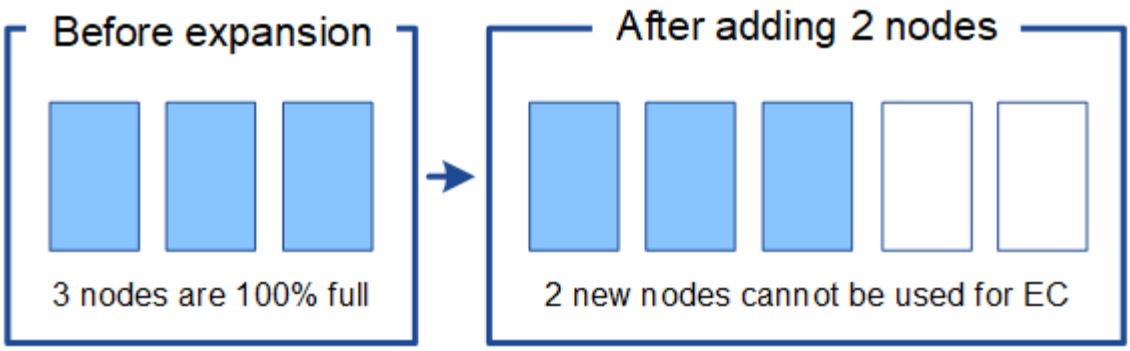
To expand the site in this example, it is recommended that you add three or more new Storage Nodes. StorageGRID requires three Storage Nodes for 2+1 erasure coding so that it can place the two data fragments and the one parity fragment on different nodes.

After you add the three Storage Nodes, the original Storage Nodes remain full, but objects can continue to be ingested into the 2+1 erasure coding scheme on the new nodes. Running the EC rebalance procedure is not recommended for this case: running the procedure will temporarily decrease performance, which might impact client operations.

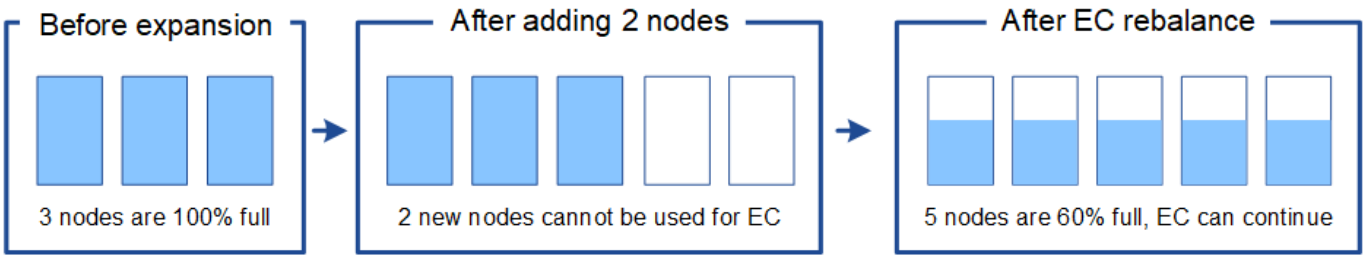


When to perform an EC rebalance

As an example of when you should perform the EC rebalance procedure, consider the same example, but assume that you can only add two Storage Nodes. Because 2+1 erasure coding requires at least three Storage Nodes, the new nodes cannot be used for erasure-coded data.



To resolve this issue and make use of the new Storage Nodes, you can run the EC rebalance procedure. When this procedure runs, StorageGRID redistributes erasure-coded data and parity fragments among all Storage Nodes at the site. In this example, when the EC rebalance procedure is complete, all five nodes are now only 60% full, and objects can continue to be ingested into the 2+1 erasure coding scheme on all Storage Nodes.



Considerations for EC rebalancing

In general, you should only run the EC rebalance procedure in limited cases. Specifically, you should perform EC rebalancing only if all of the following statements are true:

- You use erasure coding for your object data.
- The **Low Object Storage** alert has been triggered for one or more Storage Nodes at a site, indicating that the nodes are 80% or more full.
- You are unable to add the recommended number of new Storage Nodes for the erasure-coding scheme in use.

  Adding storage capacity for erasure-coded objects

- Your S3 and Swift clients can tolerate lower performance for their write and read operations while the EC rebalance procedure is running.

How the EC rebalance procedure interacts with other maintenance tasks

You cannot perform certain maintenance procedures at the same time you are running the EC rebalance procedure.

Procedure	Allowed during EC rebalance procedure?
Additional EC rebalance procedures	No. You can only run one EC rebalance procedure at a time.

Procedure	Allowed during EC rebalance procedure?
Decommission procedure EC data repair job	No. • You are prevented from starting a decommission procedure or an EC data repair while the EC rebalance procedure is running. • You are prevented from starting the EC rebalance procedure while a Storage Node decommission procedure or an EC data repair is running.
Expansion procedure	No. If you need to add new Storage Nodes in an expansion, you should wait to run the EC rebalance procedure until after you have added all new nodes. If an EC rebalance procedure is in progress when you add new Storage Nodes, data will not be moved to those nodes.
Upgrade procedure	No. If you need to upgrade StorageGRID software, you should perform the upgrade procedure before or after running the EC rebalance procedure. As required, you can terminate the EC rebalance procedure to perform a software upgrade.
Appliance node clone procedure	No. If you need to clone an appliance Storage Node, you should wait to run the EC rebalance procedure until after you have added the new node. If an EC rebalance procedure is in progress when you add new Storage Nodes, data will not be moved to those nodes.
Hotfix procedure	Yes. You can apply a StorageGRID hotfix while the EC rebalance procedure is running.
Other maintenance procedures	No. You must terminate the EC rebalance procedure before running other maintenance procedures.

How the EC rebalance procedure interacts with ILM

While the EC rebalance procedure is running, avoid making ILM changes that might change the location of existing erasure-coded objects. For example, do not start using an ILM rule that has a different Erasure Coding profile. If you need to make such ILM changes, you should abort the EC rebalance procedure.

Related information

Rebalancing erasure-coded data after adding Storage Nodes

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